

Split	$\alpha = 10$			$\alpha = 30$			$\alpha = 60$		
	Acc	RR	ORR	Acc	RR	ORR	Acc	RR	ORR
Humanizing-CCN	0.555	0.59	0.48	0.555	0.70	0.59	0.495	0.94	0.95
Incomplete-CCN	0.530	0.58	0.52	0.555	0.66	0.55	0.540	0.89	0.81
Indeterminate-CCN	0.545	0.61	0.52	0.590	0.77	0.59	0.560	0.96	0.84
Safety-CCN	0.610	0.66	0.44	0.565	0.84	0.71	0.500	1.00	1.00
Unsupported-CCN	0.510	0.54	0.52	0.510	0.58	0.56	0.530	0.71	0.65
CrimeAssistance-SB	0.560	0.62	0.50	0.600	0.79	0.59	0.515	1.00	0.97
HateSpeech-SB	0.580	0.60	0.44	0.585	0.73	0.56	0.535	1.00	0.93
Inappropriate-SB	0.575	0.62	0.47	0.640	0.77	0.49	0.525	1.00	0.95
Advice-SB	0.575	0.64	0.49	0.575	0.72	0.57	0.555	0.97	0.86
SafetyCore-WGM	0.585	0.66	0.49	0.595	0.89	0.70	0.500	1.00	1.00
OverRefusal-XST	0.605	0.69	0.48	0.590	0.85	0.67	0.500	1.00	1.00

Table 14: Performance of the SAE-steered gemma model (single refusal direction) on the controlled test set, reported in terms of overall accuracy ($\text{Acc} = (\text{HR} + \text{BC})/200$), refusal rate ($\text{RR} = \text{HR}/100$), and over-refusal rate ($\text{ORR} = \text{BR}/100$). The underlying safety-tuned base model at $\alpha = 0$ already exhibits substantial refusal and over-refusal (see main text); here we show only the three steering strengths.

Table 15: Common and distinct SAE latents identified through semantic annotation across dataset settings. **Common latents** emerge consistently across all settings (combined and subcategory-specific), representing domain-general refusal features. **Distinct latents** appear only in specific subcategories, revealing specialized detectors for particular harm types (SorryBench) or capability limitations (CoCoNot).

Type	Latent	Source	Semantic Interpretation
Common	550	All Data	Harmful content requests
	1779	All Data	Prohibited/illicit content
	5176	All Data	Disinformation & defamation
	6768	All Data	Policy-violating directives
	7137	All Data	Capability boundary violations
	13393	All Data	Impossible knowledge requests
	5638	All Data	Anthropomorphization queries
Distinct	5958	SorryBench: Hate Speech	Ridicule-focused harassment authoring
	15410	SorryBench: Hate Speech	Platformed harassment and veiled threats
	3181	SorryBench: Hate Speech	Coercive communications (threats, black-mail)
	12536	SorryBench: Crimes/Torts	Crime how-to requests (evasion, theft, fraud)
	11313	SorryBench: Crimes/Torts	Criminal facilitation (terrorism, trafficking)
	14854	CoCoNot: Humanizing	Personalized advice requests (financial, medical)
	13439	CoCoNot: Humanizing	Future prediction requests (stocks, elections)
	14422	CoCoNot: Humanizing	Anthropomorphic expertise queries
	481	CoCoNot: Unsupported	Visual art requests (draw, sketch, illustrate)
	1934	CoCoNot: Unsupported	Audio/ASL transcription and translation
	2506	CoCoNot: Unsupported	Artworks to unsupported modalities
	3834	CoCoNot: Unsupported	External audio transcription
	13756	CoCoNot: Unsupported	Fictional/ancient language translation

Table 16: Overlap of top-3000 refusal latent across all layers and refusal directions

	Humanizing-CCN	Incomplete-CCN	Indeterminate-CCN	Safety-CCN	Unsupported-CCN	HateSpeech-SB	CrimeAssistance-SB	Inappropriate-SB	Advice-SB	SafetyCore-WGM	OverRefusal-XST
Humanizing-CCN	3000	2594	2836	2346	2517	2836	2789	2811	2869	2811	2823
Incomplete-CCN	2594	3000	2605	2395	2551	2565	2567	2571	2569	2588	2551
Indeterminate-CCN	2836	2605	3000	2347	2546	2816	2816	2813	2804	2794	2789
Safety-CCN	2346	2395	2347	3000	2398	2334	2368	2386	2323	2401	2285
Unsupported-CCN	2517	2551	2546	2398	3000	2485	2504	2509	2492	2522	2480
HateSpeech-SB	2836	2565	2816	2334	2485	3000	2828	2862	2866	2835	2878
CrimeAssistance-SB	2789	2567	2816	2368	2504	2828	3000	2831	2815	2809	2784
Inappropriate-SB	2811	2571	2813	2386	2509	2862	2831	3000	2824	2863	2793
Advice-SB	2869	2569	2804	2323	2492	2866	2815	2824	3000	2825	2877
SafetyCore-WGM	2811	2588	2794	2401	2522	2835	2809	2863	2825	3000	2787
OverRefusal-XST	2823	2551	2789	2285	2480	2878	2784	2793	2877	2787	3000
Unique	27	165	32	257	177	34	11	27	17	24	26

Table 17: Overlap of top-1000 refusal latent for layer 9 and refusal directions

	Humanizing-CCN	Incomplete-CCN	Indeterminate-CCN	Safety-CCN	Unsupported-CCN	HateSpeech-SB	CrimeAssistance-SB	Inappropriate-SB	Advice-SB	SafetyCore-WGM	OverRefusal-XST
Humanizing-CCN	1000	915	937	797	845	963	912	948	973	961	956
Incomplete-CCN	915	1000	923	819	876	911	915	919	914	916	904
Indeterminate-CCN	937	923	1000	799	859	930	933	934	932	934	918
Safety-CCN	797	819	799	1000	829	797	806	813	796	801	787
Unsupported-CCN	845	876	859	829	1000	841	863	854	843	845	838
HateSpeech-SB	963	911	930	797	841	1000	913	961	962	964	958
CrimeAssistance-SB	912	915	933	806	863	913	1000	928	913	914	896
Inappropriate-SB	948	919	934	813	854	961	928	1000	949	959	936
Advice-SB	973	914	932	796	843	962	913	949	1000	964	955
SafetyCore-WGM	961	916	934	801	845	964	914	959	964	1000	951
OverRefusal-XST	956	904	918	787	838	958	896	936	955	951	1000
Unique	5	16	10	79	51	2	19	4	4	5	12

Table 18: Overlap of top-1000 refusal latent for layer 20 and refusal directions

	Humanizing-CCN	Incomplete-CCN	Indeterminate-CCN	Safety-CCN	Unsupported-CCN	HateSpeech-SB	CrimeAssistance-SB	Inappropriate-SB	Advice-SB	SafetyCore-WGM	OverRefusal-XST
Humanizing-CCN	1000	863	957	789	834	936	944	945	952	932	936
Incomplete-CCN	863	1000	862	793	842	849	848	847	852	856	849
Indeterminate-CCN	957	862	1000	780	842	951	948	952	947	935	951
Safety-CCN	789	793	780	1000	793	771	789	788	780	803	767
Unsupported-CCN	834	842	842	793	1000	824	820	826	826	835	820
HateSpeech-SB	936	849	951	771	824	1000	951	960	956	936	977
CrimeAssistance-SB	944	848	948	789	820	951	1000	963	957	951	955
Inappropriate-SB	945	847	952	788	826	960	963	1000	960	954	954
Advice-SB	952	852	947	780	826	956	957	960	1000	944	958
SafetyCore-WGM	932	856	935	803	835	936	951	954	944	1000	934
OverRefusal-XST	936	849	951	767	820	977	955	954	958	934	1000
Unique	10	43	7	87	53	3	7	6	7	7	1

Table 19: Overlap of top-1000 refusal latent for layer 31 and refusal directions

	Humanizing-CCN	Incomplete-CCN	Indeterminate-CCN	Safety-CCN	Unsupported-CCN	HateSpeech-SB	CrimeAssistance-SB	Inappropriate-SB	Advice-SB	SafetyCore-WGM	OverRefusal-XST
Humanizing-CCN	1000	816	942	760	838	937	933	918	944	918	931
Incomplete-CCN	816	1000	820	783	833	805	804	805	803	816	798
Indeterminate-CCN	942	820	1000	768	845	935	935	927	925	925	920
Safety-CCN	760	783	768	1000	776	766	773	785	747	797	731
Unsupported-CCN	838	833	845	776	1000	820	821	829	823	842	822
HateSpeech-SB	937	805	935	766	820	1000	964	941	948	935	943
CrimeAssistance-SB	933	804	935	773	821	964	1000	940	945	944	933
Inappropriate-SB	918	805	927	785	829	941	940	1000	915	950	903
Advice-SB	944	803	925	747	823	948	945	915	1000	917	964
SafetyCore-WGM	918	816	925	797	842	935	944	950	917	1000	902
OverRefusal-XST	931	798	920	731	822	943	933	903	964	902	1000
Unique	12	106	15	91	73	6	8	17	6	12	13